Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
6	(a)		$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{5730} = 1.2097 \times 10^{-4}$ [yr^{-1}] **or** 3.84×10^{-12} [s^{-1}] (1) Correct use of 70% i.e. 0.7 (1) Taking logs (1) so $t = \frac{1}{\lambda}\ln\left(\frac{N_0}{N}\right) = \frac{1}{1.2097\times10^{-4}}\ln\left(\frac{1}{0.7}\right) = 2948$ [yr] **or** 9.29×10^{10} [s] (1) Award 3 marks for 9950 years **Alternative:** Correct use of 70% i.e. 0.7 (1) Substitution e.g. $0.7 = \frac{1}{2^n}$ (1) Taking logs e.g. $\ln 0.7 = -n \ln 2$ (1) Correct answer (1)	 1 1	1 1		4	4	
	(b)		Substitution (1) $\left(\frac{A}{A_0}\right) = e^{-\left(1.2097\times10^{-4}\right)(7000)}$ = 0.429 (1) = 42.9% (1) **Alternative:** Number half-lives = $\frac{7000}{5730}$ = (1.22) (1) Substitution i.e. $A = \frac{A_0}{2^{1.22}}$ (1) Answer = 43% (1)	1	 1 1		3	3	
			Question 6 total	**3**	**4**	**0**	**7**	**7**	**0**